

Net Grid Replacement Strategy

By Team B (Aidan, Aishah, Stephen, and Cooper)

Overview

We created a comprehensive investment strategy from the seat of a large investor looking into the net new replacement capacity and timing in the energy sector for the next 20 years.

Quantitative Forecasting Model

Estimate net new power capacity needs over 20 years, factoring in demand growth, plant retirements, reserve margins, and regional trends.

Key sheets that helped us organize and process all of the data:

- **Input Sheet:** This sheet contained key parameters, such as the **annual demand growth rate** (1.00%) and the **reliability margin** (15%), which were essential in adjusting demand forecasts and determining reserve capacity.
- **Current Mix Sheet:** This sheet tracked the existing energy capacity by technology type and region, which was used as the baseline to assess future build-out needs.
- **Demand Forecast Sheet:** In this sheet, we projected regional energy demand for the next 20 years. The demand values were adjusted using the growth rate and reliability margin.
- **Retirement Forecast Sheet:** Here, we kept track of the retirements expected for each technology across regions, which was crucial for determining how much capacity would be lost each year and which technologies needed to be replaced.

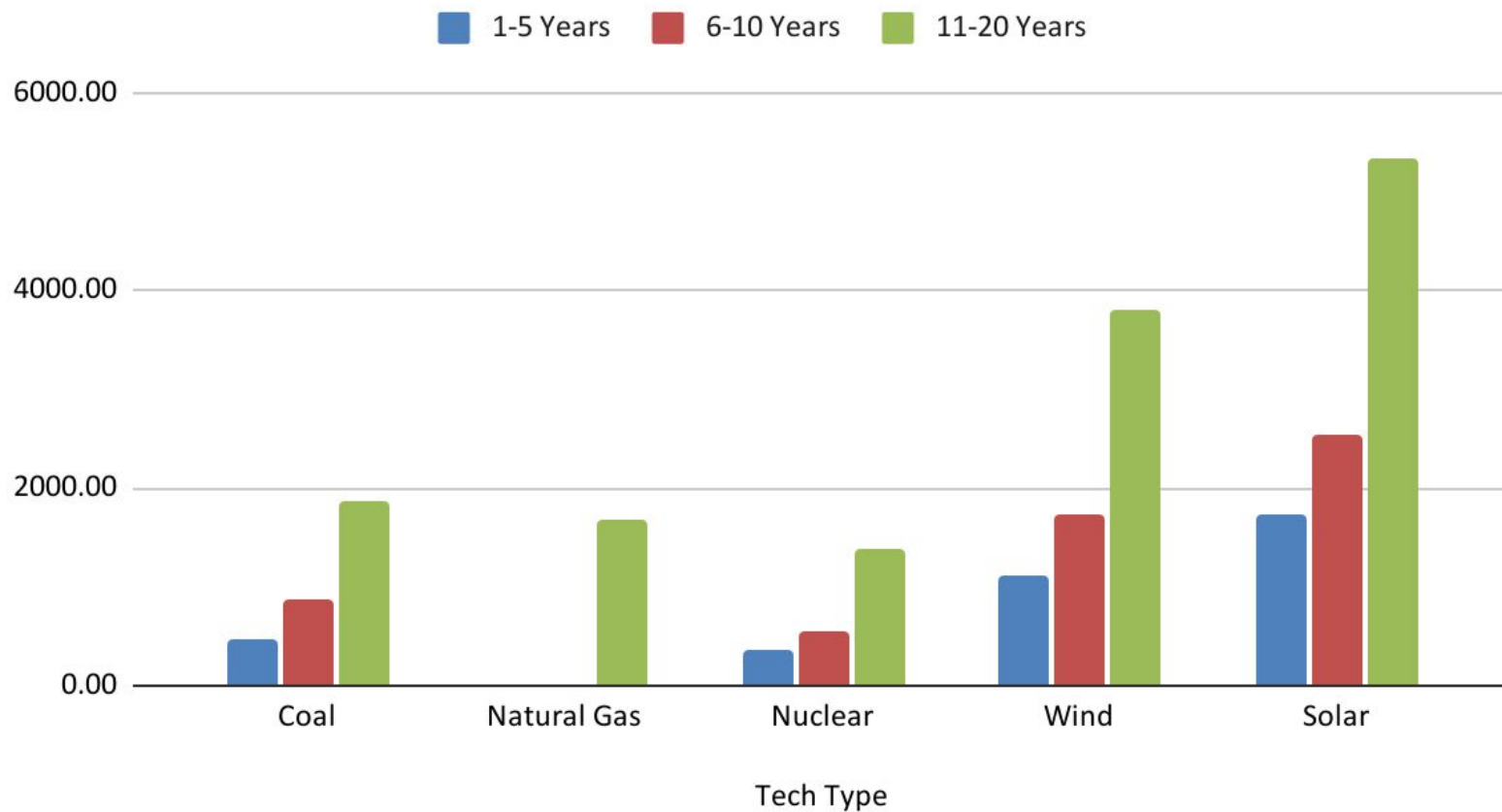
All the data used in these sheets were taken from the EIA (US Energy Information Administration) government website.

Quantitative Forecasting Model

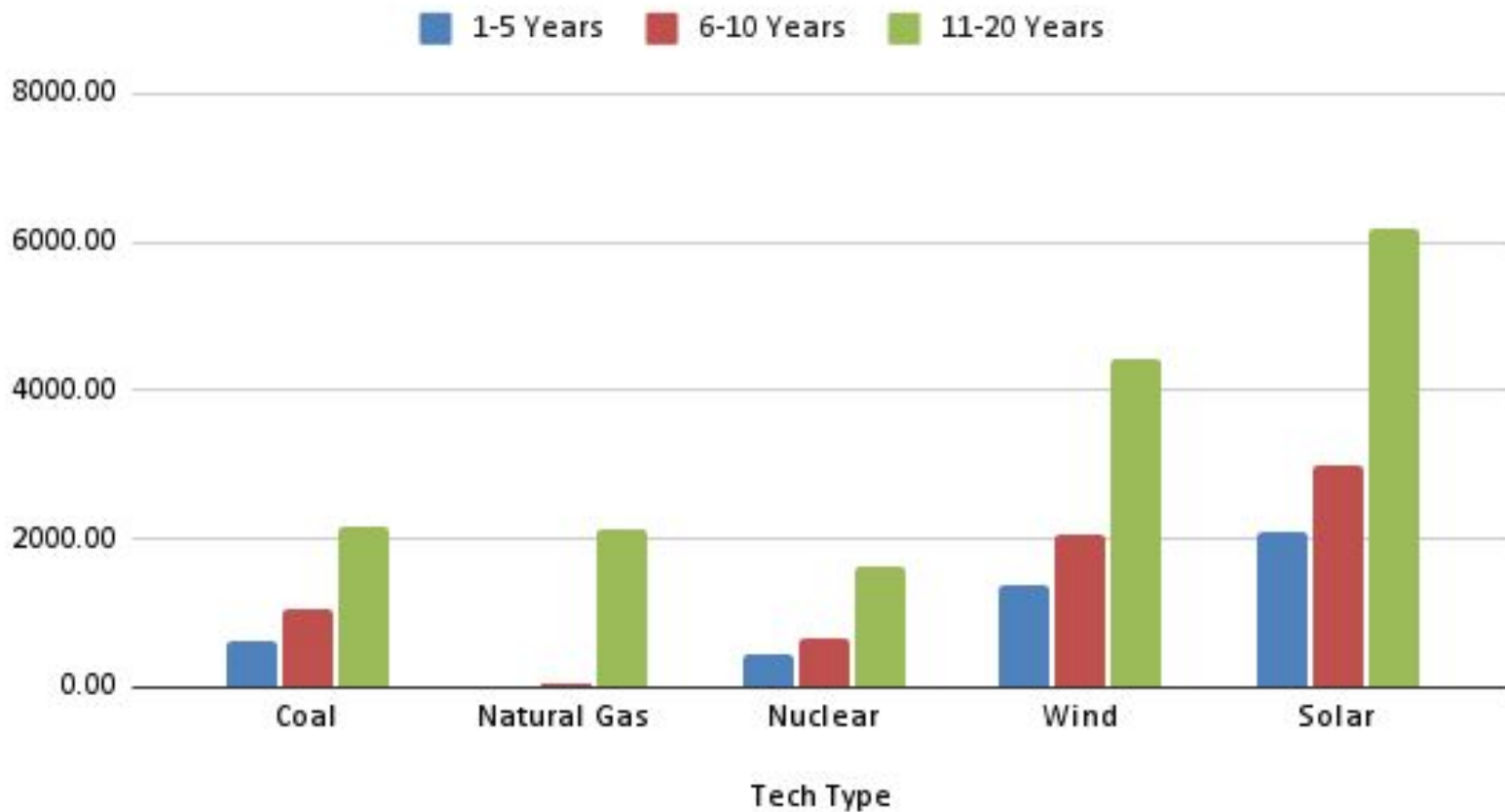
By combining the insights from the Current Mix Sheet, Demand Forecast Sheet, and Retirement Forecast Sheet, we can accurately model the net new capacity (build requirements) needed each year to meet demand, replace retiring assets, and maintain grid stability. This approach ensures that we are building enough capacity in the right places at the right times, guiding investment decisions for a reliable and sustainable energy future.

Trends in Build Requirements

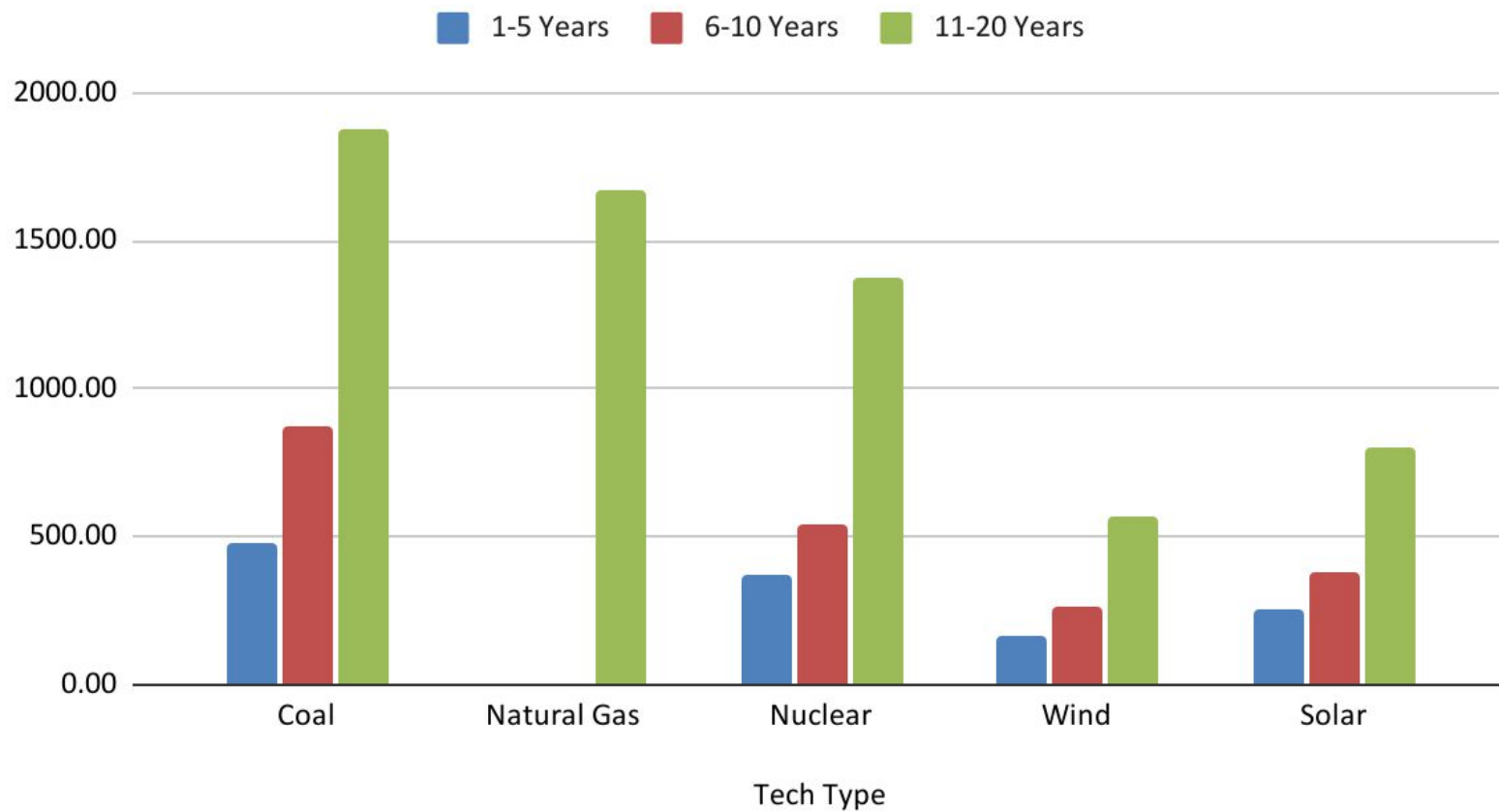
Forecast Capacity Requirements



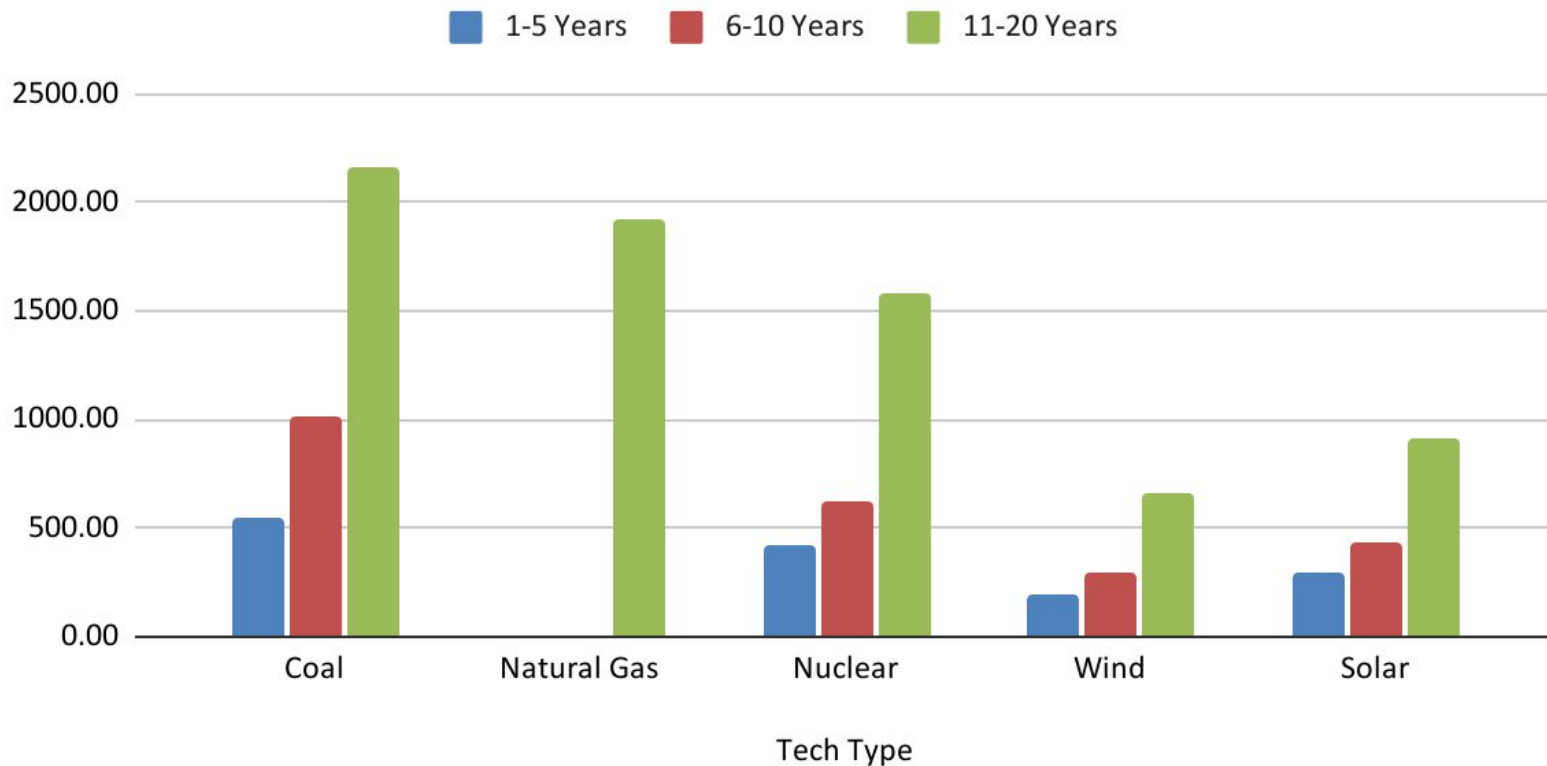
Forecast Capacity Requirements with Reserve Margin



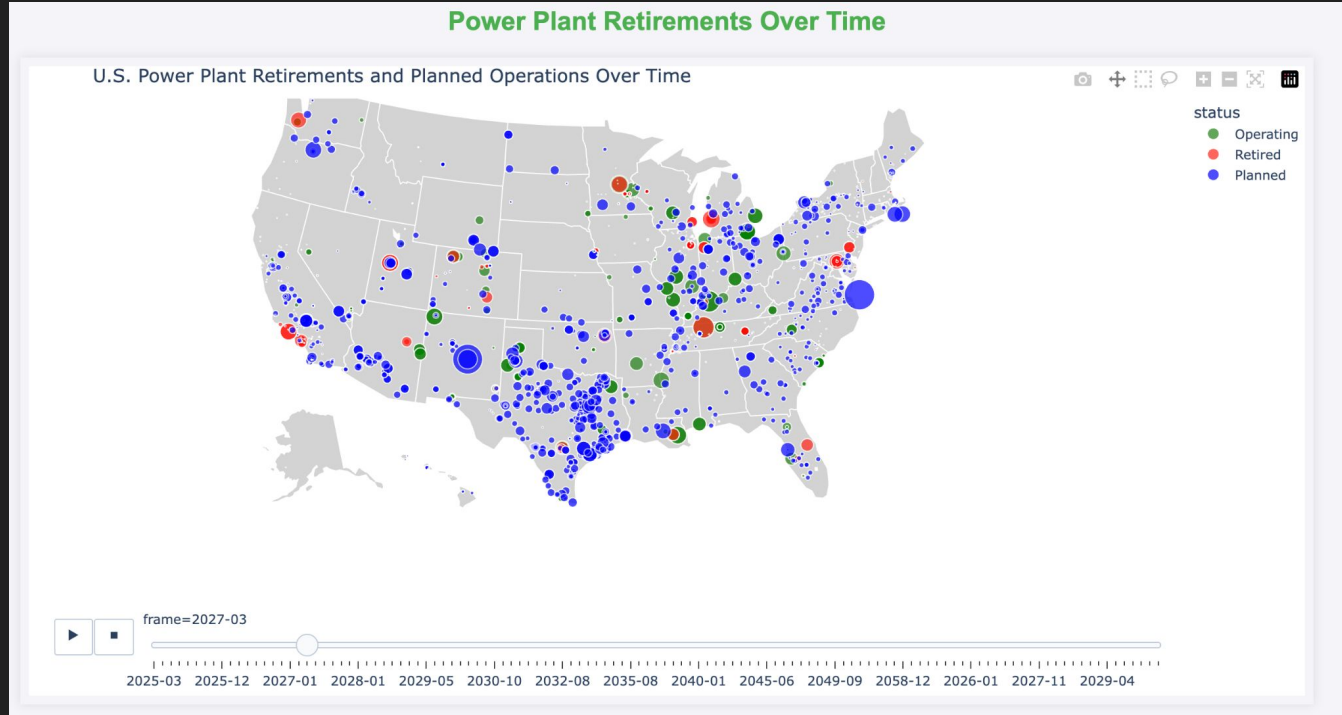
Forecast Capacity Requirements with Battery Storage



Forecast Capacity Requirements with Reserve Margin and Battery Capacity



Asset Retirement Trajectory Visualization



Major Investment Categories

- Hybrid Solar PV + Storage Systems
 - High long term profitability and benefits due to supportive regulation
 - Avoid panel and cell manufacturers due to their volatile supply chains
- Transmission-Connected Storage Portfolios
 - Focus on storage is key as the country continues to electrify
 - Transmission itself is a bad bet as a result of long waiting queues and tight regulation
- Hydrogen-Ready Natural Gas Infrastructure
 - Focused on maximized short term profitability with resilience to future climate change regulations
 - Specialized hydrogen blending equipment providers will provide better returns than general natural gas infrastructure

Portfolio Summary

Portfolio A: Transition Backbone Strategy (55% Allocation)

- Focused on utility scale generation and storage
- 9.5–12.5% ROI driven by Solar PV, Storage, and Onshore Wind
- Faces technology obsolescence and material supply risks

Portfolio B: Distributed Resilience Strategy (45% Allocation)

- Focused on distributed resources and customer level (behind-the-meter) storage
- 11–16% ROI driven by high margin consumer technology providers
- Faces local regulation and adoption challenges

Performance Tracking Dashboard

Tracks and portrays the \$10.5 billion for both portfolio A & B.

